THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 1

Chapter 2: **Electricity: Circuits and their Components** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A device that converts chemical energy into electrical energy is: 1

(a) an electric cell **(b)** a switch **(c)** a fuse **(d)** a bulb

2. Materials that allow electric current to pass through them are: 1

(a) fuses **(b)** conductors **(c)** insulators **(d)** cells

3. When current flows through a wire, the wire becomes: 1

(a) wet **(b)** hot **(c)** cold **(d)** magnetic only

4. Two or more cells joined together form a: 1

(a) fuse **(b)** switch **(c)** battery **(d)** circuit breaker

5. Materials that do not allow electric current to pass through them are: 1

(a) conductors **(b)** metals **(c)** insulators **(d)** filaments

6. The heating effect of electric current is used in: 1

(a) radio **(b)** fan only **(c)** compass **(d)** electric iron

7. The path along which electric current flows is called an: 1

(a) electric wire only **(b)** electric field
(c) electric circuit **(d)** electric switch

8. Which of these is a good conductor of electricity? 1

(a) rubber **(b)** copper **(c)** wood **(d)** plastic

9. An electric heater has a coil of: 1

(a) wood **(b)** plastic **(c)** copper wire **(d)** nichrome wire

10. A circuit in which the current flows continuously is: 1

(a) an open circuit **(b)** a broken circuit **(c)** a closed circuit **(d)** a short wire

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. What is an electric circuit? 2

12. How can you make a battery from cells? 2

13. Draw the symbols (describe) for cell and bulb. 2

14. Differentiate between an open and a closed circuit. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain why the bulb does not glow when the switch is off. 3

16. How does a fuse protect an electrical appliance? 3

17. What happens when two cells are connected in series with the same polarity? 3

OR

Explain why a bulb does not glow when a rubber piece is connected between two wires. 3

18. Why are heating coils made of nichrome? 3

OR

Explain why we should not use a damaged wire. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe an activity to show that some materials conduct electricity and others do not. Name five objects. 5

OR

What is a battery? Explain how three 1.5 V cells in series produce 4.5 V and what happens to bulb brightness. 5

20. Explain the working of a simple electric circuit and the function of cell, bulb, switch and wires. 5

OR

Compare conductors and insulators on five points with examples. 5

END OF QUESTION PAPER — Total: 40 Marks

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